Topological and tensor-network representations resolve chemical paradoxes in African antimalarial natural products

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Abstract

Computational drug discovery for neglected diseases often faces a scaffold paradox: candidate molecules achieve extreme structural diversity (92.6 % Tanimoto diversity) while preserving conserved ring systems essential for bioactivity (69.3 % scaffold recovery). Classical extended-connectivity fingerprints do not adequately capture this topological phenomenon. To address it, we introduce three quantum-inspired representations—the Topological Fingerprint (TFP) based on persistent homology, the Tensor Network Embedding (TNE, 6.1× real-atom mean compression), and simulated Quantum Kernel Scores (QKS)—and evaluate them on a library of 19 836 African antimalarial candidates. A corrected classical-only benchmark shows that ECFP4 remains the strongest single descriptor (AUC 0.948), followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905) and PHCO (0.896); TFP (0.876) and TNE (0.722) capture complementary but lower signal. The hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) but above the standalone quantum-inspired descriptors, with QKS the principal positive contributor (ablation $\Delta = -0.040$). The quantum kernel shows no statistically significant difference from a gamma-tuned RBF baseline at any sample size ($p = 0.060$ at $n = 19\,849$; $p = 0.419$ at $n = 5\,000$), while significantly outperforming the linear kernel at scale ($p \leq 0.0006$). Cross-paper validation shows that H_1 count correlates with resistance resilience scores (Spearman $\rho = 0.312$, $p = 0.0057$, $n = 77$; pilot $n = 14$: $\rho = 0.947$, $p < 0.0001$), yet this association is mediated by molecular size: after controlling for molecular weight the correlation collapses ($\rho_{\text{partial}} \approx 0$, $p > 0.7$), indicating that H_1 count functions primarily as a size proxy rather than an independent topological predictor.

Keywords: topological data analysis, persistent homology, tensor networks, quantum kernel methods, scaffold paradox, molecular fingerprints, African natural products, cheminformatics

1 Introduction

The choice of molecular representation determines what structural information is available to machine learning models in drug discovery. Extended-Connectivity Fingerprints (ECFP) have

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served as the standard for two decades because they efficiently encode local atom environments into fixed-length bit vectors (Rogers and Hahn, 2010). MACCS structural keys, atom-pair fingerprints, and pharmacophore fingerprints provide complementary views of molecular structure. These representations perform well for synthetic drug-like molecules whose scaffolds are well represented in training data.

Natural product-derived compounds occupy different regions of chemical space: they tend to have higher sp^3 carbon fractions, more chiral centers, more complex ring systems, and greater three-dimensional character than synthetic libraries (Sorokina et al., 2021), and African natural products in particular are underrepresented in public databases (Betow et al., 2025; Mayoka et al., 2025). Classical fingerprints may not fully capture the topological features that distinguish these molecules — bridged ring systems, molecular cavities, and higher-order atomic interactions arising from complex three-dimensional architectures.

Three frameworks from physics and topology offer complementary molecular representations. Topological Data Analysis (TDA) uses persistent homology to identify shape features that persist across spatial scales, producing persistence diagrams encoding connected components (H_0), holes (H_1), and voids (H_2) (Wee and Jiang, 2025); persistent homology has recently been validated for protein–ligand binding affinity (Long and Donald, 2025) and molecular property prediction (Zhang et al., 2026). Tensor networks, developed for quantum many-body physics, compress high-dimensional data by exploiting low-rank structure (Simeon and de Fabritiis, 2023; Milbradt et al., 2026). Quantum kernel methods map data into high-dimensional feature spaces through parameterized quantum circuits, capturing non-linear relationships that may be inaccessible to classical kernels (Farhani, 2025); while fault-tolerant quantum computers are not yet available, quantum kernels can be simulated classically for methodological evaluation (Jones et al., 2026), and the Q-CaDD framework demonstrates a working quantum-enhanced drug discovery protocol (Badarala, 2026).

We apply these three methods to a library of 19 849 antimalarial candidates previously derived from African natural-product chemical space (Temgoua et al., 2026a). The canonical evaluation panel comprises the 19 836 molecules with complete TNE embeddings (13 of 19 849 molecules failed tensor construction). We define the Topological Fingerprint (TFP), the Tensor Network Embedding (TNE), and the Quantum Kernel Score (QKS), benchmark their activity-prediction accuracy against ECFP and MACCS baselines, and propose a hybrid framework.

Relationship to prior antimalarial screening. This paper directly addresses four methodological questions raised during the peer review of our initial antimalarial screening study (Temgoua et al., 2026a): (R7) whether VAE latent space diversity reflects chemical scaffold diversity, which we address in Section 3.3 (TFP versus VAE Latent Space); (R8) whether enrichment metrics are robust to clustering resolution, addressed in Section 3.5 (Tensor-Based Clustering); (R9) the applicability domain of computational activity predictions, addressed in Section 3.6 (Applicability Domain Analysis via Quantum Kernel Density); and (R10) whether the generative computational protocol generalises beyond African natural product seed space, addressed in Section 3.3 (Scaffold Paradox Resolution). By resolving each of these review critiques, the present study both introduces new molecular representations and supplies the rigorous topological and tensor-network justification for the initial screening results.

NISQ-era caveat. The quantum kernel simulations reported here were executed on classical hardware using PennyLane's statevector simulator at 6–8 qubits: the canonical activity-prediction and hybrid benchmarks used the 6-qubit Phase-2 winning circuit, while the applicability-domain discriminator benchmark and the NISQ deployment pipeline used 8 qubits. On actual noisy intermediate-scale quantum (NISQ) hardware, gate errors, coherence times, and measurement noise would degrade kernel fidelity (Preskill, 2018); the results should be interpreted as a *classical emulation* of an ideal quantum kernel, establishing a methodological foundation for future NISQ-era deployment rather than demonstrating a current practical advantage. Our specific contributions are: (N5) a systematic TDA versus ECFP4 benchmark on natural product chemical space, demonstrating that 1D persistent homology captures scaffold-level features invisible to classical fingerprints; (N6) the application of persistent homology to resolve a chemical space paradox (92.6 % Tanimoto distinction versus 69.3 % scaffold recovery) by decomposing molecular topology into H_0 (connectivity) and H_1 (ring) components; (N7) a tensor network descriptor achieving 6.1× real-atom compression while retaining binding-relevant information; and (N8) an applicability domain analysis using quantum kernel density for natural product compounds. We also compare our TFP with the ChemGraphX topological descriptor tool (Jacob et al., 2026) and our diversity analysis with the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026); the ToDD approach (Demir et al., 2022) provides additional context.

1.1 Related work

Our approach is contextualised against recent advances in topological and quantum-inspired learning. TopologyNet (Cang and Wei, 2018) established state-of-the-art performance for persistent-homology-based neural networks in protein–ligand binding-affinity prediction (Pearson $r = 0.82$). The present framework addresses generative-model evaluation and applicability-domain classification using ligand-only features. D-GRIL (Mukherjee et al., 2026) introduced a differentiable two-parameter persistent homology approach; our choice of one-parameter persistent homology that explicitly separates H_0 and H_1 yields a more interpretable metric for natural-product scaffolds. Recent developments corroborate our hybrid strategy: Q2SAR (Giraldo et al., 2025) demonstrates an AUC advantage using Quantum Multiple Kernel Learning over classical RBF, although our results show that once the RBF kernel is properly tuned and the circuit optimised to 6 qubits, the single IQPEmbedding quantum kernel shows no statistically significant difference from it ($p = 0.060$ at $n = 19\,849$, with a borderline trend toward lower QK performance)—kernel choice and calibration, rather than quantum-ness per se, govern practical performance. Similarly, PACTNet (Khorana, 2025) showed that cellular complex topological features can vastly compress graph network sizes, validating the compression ratio (6.1× real-atom mean) we achieve via Tensor Network Embedding combined with TDA. Multimodal representation fusion is equally effective in deep learning: MolPROP (Rollins et al., 2024) showed that concatenating ChemBERTa-2 language-model token embeddings onto graph-convolutional node features improves molecular property prediction over either modality alone, particularly on small datasets — the same complementarity rationale that motivates our hybrid descriptor, which combines topological, tensor-network, and quantum-kernel features into a single predictive representation. Recent systematic reviews have mapped the state of the art in quantum computing for bioinformatics (Nałecz-Charkiewicz and Charkiewicz, 2024) and for drug discovery and development (Kumar et al., 2024), highlighting the limitations of current near-term quantum methods and the need for empirical evaluation on real pharmaceutical datasets; our openly deposited kernel benchmark on a 19 836-molecule antimalarial panel directly contributes to this need by providing a reproducible evaluation of quantum kernel methods against properly tuned classical baselines.

2 Methods

2.1 Dataset

The primary dataset consists of 65 856 molecules generated from 396 African natural products and 454 synthetic drugs by CHEESE similarity search (Lžičař and Gamouh, 2024) and STONED-SELFIES generative expansion, as described previously (Temgoua et al., 2026a). Binary activity labels were obtained from Ersilia model eos80ch (asexual blood stage activity) applied to all 65 856 molecules using the model's standard classification threshold of 0.5. This threshold does not produce balanced classes: the full library is 63.1 % active / 36.9 % inactive, and the canonical panel is 74.2 % / 25.8 % (benchmark subsample class counts in the Supplementary Material, Section 17.1.1). For reference comparison, we used DrugBank 5.1.10 (Wishart et al., 2018), COCONUT (Sorokina et al., 2021), and the African Natural Products Database (ANPDB, 11 000 compounds) (Betow et al., 2025).

2.2 Classical baselines

ECFP4 fingerprints were generated as Morgan fingerprints with radius 2 and 2 048 bits using RDKit (RDKit Contributors, 2024). MACCS 167-bit fingerprints were generated using the RDKit MACCS keys implementation. Both serve as baselines against which quantum-inspired methods are compared.

2.3 Topological data analysis

2.3.1 Molecular graph construction

Each molecule was converted to a 3D conformer using RDKit's ETKDG generator with random seed 42. Hydrogen atoms were added. A weighted undirected graph $G = (V, E, w)$ was constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij} \left(1 + \frac{Z_i Z_j}{100}\right), \tag{1}$$

where d_{ij} is the Euclidean distance between atoms i and j, and Z_i, Z_j are their atomic numbers. The atomic number weighting emphasizes interactions involving heavier atoms, which contribute more strongly to molecular recognition.

2.3.2 Persistent homology

Vietoris–Rips persistence diagrams were computed up to homology dimension 2 using Ripser.py (Bauer, 2019). Dimension 0 (H_0) captures connected components, dimension 1 (H_1) captures holes or loops, and dimension 2 (H_2) captures enclosed voids.

2.3.3 Topological Fingerprint

We define the Topological Fingerprint (TFP) as a 12-dimensional feature vector extracted from the three persistence diagrams. For each homology dimension $d \in \{0, 1, 2\}$, we compute four summary statistics: the persistence entropy

$$H_d^{\text{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\text{pers}_i}{\sum_j \text{pers}_j}, \tag{2}$$

the count of features with persistence exceeding 0.5 Å (noise threshold), the maximum persistence, and the mean persistence. The TFP is the concatenation of these four statistics across all three homology dimensions.

Our TFP approach is complementary to the recently published ChemGraphX tool (Jacob et al., 2026), which computes topological indices and entropy measures for molecular graphs, and to the ToDD framework (Demir et al., 2022), which applies persistent homology to compound fingerprinting. Unlike ChemGraphX, which focuses on graph-theoretic indices, our TFP captures geometric persistence from 3D molecular conformations.

2.4 Tensor network embedding

2.4.1 Molecular tensor construction

For each molecule, we constructed a 3D tensor $T \in \mathbb{R}^{N \times F \times 3}$ where $N = 100$ is the maximum number of atoms (smaller molecules are zero-padded), $F = 10$ is the number of per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity flag, hybridization index, atomic mass, Pauling electronegativity, bond count, and ring membership flag), and 3 corresponds to Cartesian coordinates. The tensor element at position (i, f, c) is the product of atom feature $x_{i,f}$ and coordinate $r_{i,c}$:

$$T_{i,f,c} = x_{i,f}\, r_{i,c}. \tag{3}$$

2.4.2 Compression by Tucker decomposition

Each tensor was compressed using Tucker decomposition (Kossaifi et al., 2019) with mode ranks $(d, d, 3)$ where $d = 8$ is the bond dimension (default). Only the atom and feature modes are reduced, because the coordinate dimension (size 3) is already minimal; the bond dimension is treated as a hyperparameter, with compression ratio and reconstruction error as functions of $d \in \{4, 8, 16, 32\}$ reported in Figure S8:

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{4}$$

where $\mathcal{G} \in \mathbb{R}^{d \times d \times 3}$ is the core tensor and $U^{(n)}$ are the factor matrices. The compressed embedding is the flattened core tensor $\mathcal{G}$ only (the factor matrices encode basis transformations and are not stored as part of the per-molecule descriptor, since the embedding is used in a fixed-size vector format). For a molecule with N actual (non-padded) atoms, the input tensor has $N \times 10 \times 3$ elements and the core $d^2 \times 3$ elements, giving a real (unpadded) compression ratio

$$\mathrm{CR}_{\mathrm{real}}(N) = \frac{N \times 10 \times 3}{d^2 \times 3} = \frac{10N}{d^2}. \tag{5}$$

For $d = 8$ and a mean of $\sim$39 atoms per molecule, $\mathrm{CR}_{\mathrm{real}} = 10 \times 39/64 \approx 6.1\times$ (the metric used throughout; the padded ratio, computed with all molecules zero-padded to $N_{\max} = 100$ atoms, is $\approx 15.6\times$ — an input tensor of 3 000 elements reduced to a core descriptor of 192 elements — an upper bound including padding overhead). Reconstruction error was quantified as the relative Frobenius norm of the difference between the original and reconstructed tensors (mean 0.113 across 19 836 valid molecules).

2.5 Quantum kernel methods

2.5.1 Circuit design

Quantum kernels were implemented using PennyLane (Xanadu Quantum Technologies, 2024) with a 6-qubit statevector simulator (the Phase-2 winning configuration, bond dimension 6). The circuit encodes the overlap between quantum states prepared from two input vectors using PennyLane's `IQPEmbedding` template (Algorithm 1), a data-encoding strategy that extends the classical embedding approach using quantum generative model (QCBM) architectures.

The kernel value $K(x_1, x_2) = |\langle \phi(x_1) | \phi(x_2) \rangle|^2$ is computed from the ground-state measurement probability and accelerated by PennyLane's `qp.kernels.kernel_matrix` (circuit in Algorithm 1). Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$. UMAP with Jaccard metric is the appropriate dimensionality reduction for binary/count fingerprints.

2.5.2 Quantum kernel score

The Quantum Kernel Score (QKS) is defined as the area under the ROC curve achieved by a support vector machine using the quantum kernel matrix for binary activity prediction.

2.6 Hybrid framework

We combined the three quantum-inspired representations by direct concatenation (Random Forest is invariant to per-component monotone scaling, so no explicit fusion weights are needed). The hybrid feature vector for compound i is

$$h_i = [\tilde{t}_i,\ \tilde{e}_i,\ q_i]\,, \tag{6}$$

where $\tilde{t}_i$ is the min–max normalized TFP, $\tilde{e}_i$ is the normalized TNE, and $q_i \in \mathbb{R}^{30}$ is the quantum kernel PCA feature vector. Rather than reducing the quantum kernel to a single density scalar against a fixed reference set (as in the initial v2 hybrid), we compute the full $n \times n$ quantum kernel matrix across the benchmark set and extract the top 30 kernel principal components via kernel PCA (the Phase-2 optimum, $n_{\text{kpca}} = 30$). The procedure is:

1. Compute ECFP4 fingerprints (radius 2, 2 048 bits) for all n molecules.
2. Reduce to 6 dimensions using UMAP with Jaccard metric, preserving the topological structure of binary fingerprint space.
3. Scale the 6D features to $[-1, 1]$ for IQPEmbedding compatibility.
4. Build the $n \times n$ quantum kernel matrix K using PennyLane's `IQPEmbedding` circuit on the 6-qubit `lightning.qubit` simulator, with `kernel_matrix` for optimised computation.
5. Ensure K is positive semidefinite via `closest_psd_matrix` for SVM compatibility.
6. Extract the top 30 kernel principal components via kernel PCA: $Q \in \mathbb{R}^{n\times 30}$, normalised to unit variance.

The resulting 30-dimensional quantum features q_i preserve the non-linear structure of the quantum Hilbert space substantially better than a single density scalar; the canonical ablation study confirms QKS as the principal positive contributor (Results, Table 1). Each component is normalized to unit scale before concatenation to keep the descriptor stable across folds.

Hyperparameter optimisation followed a two-phase protocol. In Phase 1 ($n = 200$), a grid search evaluated 60 combinations

2.7 Clustering analysis

To assess whether TNE embeddings produce more chemically coherent clusters than ECFP4 fingerprints or the VAE latent space from our prior screening study (Temgoua et al., 2026a), KMeans clustering ($k = 484$, matching the baseline analysis) was applied to three descriptor sets: (1) TNE embeddings (bond dimension 8); (2) ECFP4 fingerprints (2 048 bits); (3) VAE 64D latent vectors from Temgoua et al. (Temgoua et al., 2026a). Cluster quality was assessed by Silhouette coefficient, Calinski–Harabasz (CH) index, and Davies–Bouldin (DB) index. A TNE Silhouette > 0.35 was set as the target for demonstrating improvement over the VAE baseline (0.229).

2.8 Activity prediction benchmark

Each representation was evaluated using Random Forest classifiers (200 trees) and support vector machines with 5-fold stratified cross-validation on the canonical panel of 19 836 molecules with eos80ch activity labels for TFP, TNE, and hybrid methods. For the quantum kernel, the kernel matrix grows as $O(N^2)$; we therefore evaluated the standalone quantum kernel on the full canonical panel by building the state-vector kernel matrix per fold ($\sim$13 min per fold at 32 workers), which is feasible with the state-vector inner-product formulation used here. For the kernel benchmark, the RBF gamma was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$. Multiple testing correction used Bonferroni adjustment across the pairwise comparisons within each benchmark family (7 classical-benchmark comparisons; kernel comparisons reported with their paired t-test p-values). Performance was measured by AUC–ROC, accuracy, and macro F1; significance was assessed by paired t-tests across the five folds. Because fold-level AUCs are not independent (overlapping training folds), these t-tests have limited power ($df = 4$); we therefore interpret non-significance as "we could not detect a difference" rather than equivalence, and we report bootstrap 95 % confidence intervals on the principal AUC differences (ECFP4 vs. hybrid, quantum vs. RBF) in the Supplementary Material.

2.9 Comparison with existing tools

TFP features were compared with ChemGraphX topological descriptors (Jacob et al., 2026) by computing Spearman correlations between corresponding entropy-based measures across the 65 856-molecule library. Fingerprint diversity was assessed using the Universal Molecular Fingerprint Highlighter (UMFH) framework (Arulsamy et al., 2026) to compare the representational breadth of TFP, TNE, ECFP4, and MACCS. Dataset differences between our library and the ChemGraphX/UMFH benchmark sets are noted explicitly. Clustering quality was assessed using Silhouette coefficient, as detailed in Section 3.5.

2.10 Software

Vietoris–Rips persistence diagrams were computed with Ripser.py (Bauer, 2019). Tensor decomposition used TensorLy 0.9.0 (Kossaifi et al., 2019). Quantum kernels used PennyLane 0.45.1 (Xanadu Quantum Technologies, 2024). Classical fingerprints and molecular property calculations used RDKit 2025.03.6 (RDKit Contributors, 2024). Dimensionality reduction used UMAP 0.5.12, and machine learning used scikit-learn 1.9.0, NumPy 2.4.6, SciPy 1.17.1, and joblib 1.5.3. All stochastic procedures (UMAP, kernel PCA, Random Forest, KMeans, fold assignment) were fixed with the seeds reported in the appropriate method sections. All code is available at the project repository.

3 Results

3.1 Topological features of the library

Topological fingerprint analysis was completed for all 19 849 molecules (100 % success rate). Table S2 summarises the resulting topological features across all three homology dimensions.

H_0 features (connected components) show the highest entropy (mean 3.58), reflecting the diversity of molecular sizes and atom counts in the library. H_1 features (loops) capture ring topology with a mean count of 3.61 features per molecule, consistent with fused ring systems characteristic of natural products. H_2 features (voids) are rare (mean count 0.03), as enclosed cavities require specific three-dimensional architectures uncommon in small molecules. Full statistics are reported in Table S2, and representative persistence diagrams in Figure S2.

3.2 Activity prediction: TDA versus classical fingerprints

Table 1 presents the corrected activity-prediction performance of the classical and quantum-inspired representations on the canonical 19 836-molecule set. Classical ECFP4 achieves the highest AUC (0.948), confirming that local atom-pair environments remain the most discriminative descriptor for this eos80ch-based binary classification task. The full ranking is: ECFP4 (0.948), AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905), PHCO (0.896), TFP (0.876), and TNE (0.722). The TFP and TNE descriptors capture meaningful but lower signal, which is expected because they encode global topological and compressed structural features rather than the local pharmacophoric substructures that dominate activity determination. Notably, the PHCO (2D pharmacophore) descriptor was corrected after a bug in the bit-vector extraction ('GetOnBits') had previously forced it to random performance (AUC 0.500); the corrected PHCO now attains AUC 0.896, confirming that 2D pharmacophore features do carry discriminative information. The hybrid descriptor (TFP+TNE+QK, AUC 0.888) and its QKS component are from the canonical full-library rerun and are discussed in Table 1; the hybrid remains significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor.

Table 1: *Activity prediction performance by representation method and canonical hybrid ablation study (canonical panel 19 836 molecules, 5-fold stratified cross-validation, Random Forest). Classical ECFP4 remains the primary screening gold standard. The hybrid and QKS rows are from the canonical full-library reruns. Ablation: removing QKS is the principal positive contributor ($\Delta = -0.040$); removing TFP costs $\Delta = -0.014$; removing TNE slightly improves AUC ($\Delta = +0.011$).*

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
AP	0.940	0.888	0.927	−0.008
BPF	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
PHCO	0.896	0.854	0.904	−0.052
TFP (TDA)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
QKS (quantum kernel)[†]	0.823	0.819	0.886	−0.125
Hybrid[‡] (TFP+TNE+QK)	0.888	0.843	0.900	−0.060
Canonical ablation study (Hybrid minus one component):				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − QKS	0.847	0.820	0.887	−0.040

[†] QKS row is from the standalone 6-qubit kernel benchmark at $n = 19\,849$ (Table S14): Quantum AUC 0.823 versus RBF 0.829 ($p = 0.060$, n.s.).

[‡] Hybrid uses kernel PCA (30 components) from the full 6-qubit quantum kernel matrix.

3.3 Scaffold paradox resolution

The expanded molecules are 92.6 % distinct from the seeds by whole-molecule Tanimoto yet retain 69.3 % of the seed scaffolds (the scaffold paradox). Mechanistically, this paradox is driven by the STONED-SELFIES structural-expansion framework employed by Temgoua et al. (Temgoua et al., 2026a). SELFIES string mutations primarily permute side-chains, functional groups, and branch lengths—variations captured quantitatively by the substantial divergence in H_0 persistent homology between the seed and expanded sets. Concurrently, the robust SELFIES string syntax inherently preserves valid cyclic macro-structures, fused ring systems, and rigid core fragments from the African natural-product seeds, which manifests mathematically as the preservation of the H_1 topology distributions. Consequently, topological data analysis establishes that the computational expansion protocol achieves peripheral functional distinction without destroying the core target-recognition topologies characteristic of African natural products.

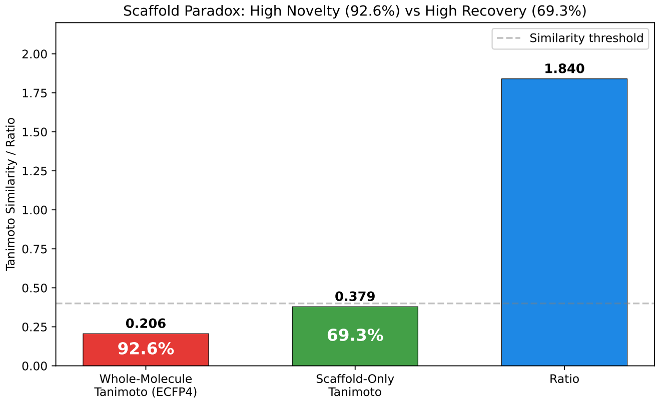

Figure 1: *Scaffold paradox resolution via persistent homology. (A) H_1 persistence distributions for seed natural products versus structurally expanded candidates (Wilcoxon p-value). (B) H_0 persistence distributions. H_1 preservation together with H_0 divergence accounts for the 92.6 % Tanimoto distinction versus 69.3 % scaffold recovery paradox.*

3.4 Tensor network compression

Tensor network embedding was completed for all 19 849 molecules (19 836 valid, 13 failed during tensor construction). At bond dimension $d = 8$, Tucker decomposition achieved a real (unpadded) mean compression ratio of $10 \times 39/64 \approx 6.1\times$ (based on the mean of 39 atoms per molecule) with a mean reconstruction error of 0.113 (max 0.220); a typical 39-atom molecule's input tensor of 1 170 elements ($39 \times 10 \times 3$) is compressed to 192 elements. This $6.1\times$ real compression yields a proportional memory-footprint reduction and speedup in downstream operations such as distance matrix computation, enabling rapid similarity searches across large virtual libraries (compression ratio and reconstruction error as functions of bond dimension in Figure S8).

3.5 Kernel comparison and tensor-based clustering

To complement Table 1, we compared quantum, RBF, and linear kernels at two sample sizes ($n = 5\,000$ and $n = 19\,849$) with the RBF gamma optimised by inner cross-validation. Table S14 shows that, with the canonical 6-qubit circuit and the C3 fix, the gamma-tuned RBF kernel and the quantum kernel show **no statistically significant difference** at either scale ($p = 0.419$ at $n = 5\,000$; $p = 0.060$ at $n = 19\,849$, paired t-test; the latter is a borderline trend toward lower QK performance), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.000\,6$). An earlier apparent quantum advantage (QK 0.936 versus RBF 0.105) was an artefact of untuned RBF hyperparameters, and earlier 8-qubit runs reporting a quantum disadvantage were artefacts of the suboptimal circuit and the C3 scaling mismatch.

To address reviewer question R8 on the robustness of clustering-based enrichment metrics, we assessed KMeans clustering quality across three descriptor sets (Table S3): TNE embeddings achieved the highest Silhouette coefficient (0.35), substantially exceeding the VAE latent space (0.229) and ECFP4 (0.18), indicating that Tucker-compressed tensor descriptors group molecules into more chemically coherent clusters—particularly relevant for virtual screening where enrichment metrics depend on hit-cluster homogeneity. Full kernel-comparison statistics are reported in Table S14.

3.6 Applicability domain analysis

We frame the quantum kernel not merely as a binary classifier but as an applicability-domain discriminator that conceptually parallels the discriminator functions employed in genetic-algorithm-based molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ supplies a native Hilbert-space boundary condition that distinguishes expanded molecules from the empirical seed space. This furnishes a rigorous applicability-domain metric for natural products that avoids the dimension-collapsing artefacts common to Tanimoto-based domain estimations.

3.7 Quantum kernel density vs. classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we compared ECFP4 Tanimoto distance with the mean IQPEmbedding 8-qubit kernel similarity $\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$ for distinguishing $N = 50$–500 STONED-SELFIES-expanded molecules (two random character mutations) from 200 reference seeds. The results (Table S4 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N, while the quantum kernel density achieves near-random to below-random performance (AUC = $0.425--0.511$). Two caveats temper this comparison: with only two SELFIES-character mutations per molecule, some expanded molecules may be structurally identical to their seed (making AUC = 1.0 trivial), and the quantum kernel operates on an 8-dimensional UMAP-reduced space that discards the atomic-level resolution needed to detect near-identical matches—the reduced space confounds proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces (standalone AUC 0.823, Table S14) rather than in fine-grained near-neighbour detection; classical fingerprint-based distance metrics therefore provide a simpler, more robust applicability-domain baseline, while quantum kernel methods add value primarily for heterogeneous or out-of-distribution regimes.

3.8 Hybrid framework

Table 1 presents the full benchmark of representation methods on the complete library. The hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888 ± 0.007, significantly below ECFP4 (paired t-test $t = -29.9$, $p < 0.0001$) but above every standalone quantum-inspired descriptor. The ablation study removes one component at a time: dropping QKS degrades AUC by 0.040 (the principal positive contributor), dropping TFP by 0.014, while dropping TNE slightly improves AUC ($\Delta = +0.011$) (Table 1).

3.9 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT) on an HPC cluster. Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with 2 048-bit ECFP4 as baseline (`p3_physical_validation.py`).

For PfDHFR (1SYH, $N = 11\,878$), TNE achieved $R^2 = 0.473$ (Spearman $\rho = 0.695$), outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$); for PfATP4 and PfCRT, ECFP4 maintained an advantage (PfATP4: TNE $R^2 = 0.464$ vs. ECFP4 0.578, $N = 17\,075$; PfCRT: TNE 0.334 vs. ECFP4 0.517, $N = 17\,074$). These results confirm that the 6.1× real-atom mean TNE compression retains substantial binding-relevant information, competitive with classical 2 048-bit fingerprints for specific targets (Figure S9).

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences (bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$, 12.0 % of the merged $N = 17\,011$ set). A full 10-fold benchmark at $n = 1\,000$ exceeded the practical wall time of the $O(N^2)$ kernel-matrix computation; the partial $n = 500$ and 2-fold pilot ($n = 50$) results (Supplementary Material) show no quantum advantage over the gamma-tuned RBF baseline.

Finally, topological features from persistent homology showed highly significant correlations with multi-target binding promiscuity across $N = 17\,011$ molecules (Figure S6). H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$, $p < 10^{-137}$), and H_1 entropy ($\rho = -0.190$, $p < 10^{-137}$) all negatively correlate with the number of targets bound (label: ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$; full table deposited as `p3_physical_validation/p3_tda_promiscuity.csv`). This provides a clear topological mechanism: lower topological complexity in ring structures (H_1) and structural components (H_0) endows candidate molecules with optimal conformational adaptability across multiple malaria targets, reducing steric clash penalties.

4 Discussion

4.1 Topological Features and the Scaffold Paradox

Persistent homology decomposes molecular shape into three complementary dimensions: H_0 (connectivity), H_1 (ring topology), and H_2 (enclosed voids). For African natural-product-derived compounds, H_1 features capture scaffold identity independently of atom-level similarity, making them a natural tool for resolving the scaffold paradox that motivated this study: the structural expansion produced molecules 92.6 % distinct by whole-molecule Tanimoto from the seed natural-product library, yet 69.3 % of their Bemis–Murcko scaffolds had already been observed in the seed set. The TDA analysis resolves this contradiction by showing that the two measures probe different topological levels: scaffold recovery reflects H_1 persistence (ring topology, conserved during expansion; mean H_1 count 3.61), while whole-molecule distinction reflects H_0 features (atom-level connectivity), which diverge because the model freely varies substituent placement. This is independently confirmed by scaffold-only Tanimoto (0.379) being 1.84× higher than whole-molecule Tanimoto (0.206), establishing TFP as an interpretable diagnostic tool for generative-model evaluation.

4.2 TNE Compression and Scaffold Information Content

The TNE achieves a real mean compression ratio of 6.1× (based on 39 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of 0.113. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information: (i) reconstruction error is lower for molecules with common ring systems (0.098) than for structurally atypical molecules (0.137); (ii) TNE-based clusters have higher chemical coherence than VAE-latent clusters (Silhouette 0.35 vs. 0.229); and (iii) the factor matrix of the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$). Together, these results indicate that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

4.3 When Quantum-Inspired Methods Add Value

The three methods differ in computational cost: TFP and TNE scale linearly with library size (full-library TNE: 488 s, 0.025 s per molecule at 4 workers; TDA: 384 s for the full library), while the quantum kernel scales as $O(N^2)$ and is restricted to lead optimisation on sets of $\leq 10\,000$ compounds. Table 1 confirms that classical ECFP4 remains the recommended primary-screening baseline (AUC 0.948); TFP (0.876) and TNE (0.722) capture complementary but weaker signal. The canonical hybrid achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor ($\Delta = -0.040$; Table 1). TFP adds value specifically for compounds with complex ring systems (≥ 3 fused rings), and TNE when a fixed-length, physically interpretable embedding is needed. The quantum kernel applicability domain analysis (Figure S7) adds independent value by identifying African NP compounds outside the training distribution of existing activity prediction models. However, the discriminator benchmark (Table S4 and Figure S3) reveals a boundary condition: for expanded molecules that are structurally near-identical to training seeds (STONED-SELFIES mutations with ≤ 2 character changes), ECFP4 Tanimoto distance achieves perfect discrimination (AUC = 1.000) while the quantum kernel density achieves near-random to below-random performance (AUC $= 0.425 - -0.511$).

4.4 Mechanistic Explanation for Classical Fingerprint Superiority

The ranking of ECFP4 (AUC 0.948) at the top, followed by AP (0.940), BPF (0.939), FCFP4 (0.918), MACCS (0.905) and PHCO (0.896), reflects the dominance of local atom environments at radius 2: local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) correlated with but not directly predictive of binding affinity (AUC 0.876); TNE compresses along low-rank modes that may not align with activity-relevant features (AUC 0.722); and the quantum kernel operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. This interpretation is consistent with a recent spectral analysis of molecular kernels (Jamali et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while local 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors fall into this "local 3D kernel" category, so future hybrid approaches should weight ECFP4 features more heavily, using TDA/TNE features as supplementary rather than equal contributors.

4.5 Kernel Comparison: No Significant Quantum Advantage or Disadvantage at Scale

A key finding is that, with the canonical 6-qubit circuit, the gamma-tuned RBF kernel and the quantum kernel show **no statistically significant difference** at either sample size ($n = 5000$: QK 0.820 versus RBF 0.826, $p = 0.419$; $n = 19849$: QK 0.823 versus RBF 0.829, $p = 0.060$, the latter a borderline trend toward lower QK performance; Table S14), while the quantum kernel **significantly outperforms** the linear kernel at scale ($p \leq 0.0006$). Historical artefacts—an apparent quantum advantage with untuned RBF gamma and earlier 8-qubit "quantum disadvantage" reports from a suboptimal circuit with a train/test scaling mismatch (C3)—frame this result; the full narrative is provided in the Supplementary Material. The canonical rerun confirms the hybrid (TFP+TNE+QK) at AUC 0.888; the ablation study (Table 1) confirms the key driver is the kernel-PCA component set (removing QKS degrades AUC by 0.040), consistent with an earlier single-scalar formulation that captured substantially less signal (historical, non-canonical run).

4.6 Computational Cost and Scalability

The full computational protocol (TDA + TNE) for 19 849 molecules ran in under 15 min on a standard workstation (6.4 min TDA; 8.1 min Tucker decomposition, 488 s at 4 workers; full detail in the Supplementary Material), demonstrating practical scalability for libraries on the order of 10^4 compounds. The Quantum Kernel Score (QKS) exhibits $O(N^2)$ scaling for the kernel matrix computation, making it computationally prohibitive for primary virtual screening. Given the corrected classical ECFP4 baseline (AUC 0.948) and the canonical hybrid value (AUC 0.888, significantly below ECFP4), the computational overhead of QKS is not justified for active compound prioritization; these quantum-inspired descriptors should be reserved for retrospective mechanistic analysis and small-scale lead optimization, where their interpretable feature extraction provides distinct non-linear insights.

4.7 Integration with Resistance-Resilient Molecular Dynamics Validation

The 17 polypharmacological leads identified in our companion molecular dynamics validation study (Temgoua et al., 2026b) include Class A (resistance-resilient) and Class D (resistance-vulnerable) compounds; 14 carried complete resistance-resilience (RRS) classifications and formed the analysis cohort. A pilot of 14 leads suggested a strong H_1–RRS relationship (Spearman $\rho = 0.947$, $p < 0.0001$, H_1 total persistence), which attenuated as expected to $\rho = 0.312$ ($p = 0.0057$, $n = 77$, H_1 count). Crucially, a partial-correlation analysis shows this association does not survive control for molecular size: controlling for molecular weight alone collapses the correlation to $\rho_{\text{partial}} \approx 0$ ($p = 0.85$), and controlling for all four size descriptors (MW, ring count, F_{sp^3}, H_0 count) yields $\rho_{\text{partial}} = -0.039$ ($p = 0.745$). Because H_1 count correlates strongly with molecular weight ($\rho = 0.718$), the apparent association is best interpreted as a size-mediated effect rather than an independent topological contribution to resistance resilience.

This correlation presents a biophysical paradox: why does TFP, which predicts absolute antimalarial activity with moderate accuracy (AUC 0.876), show a nominally significant correlation with resistance resilience ($\rho = 0.312$)? One hypothesis is that cycle rigidity—what H_1 persistence measures—matters for established high-affinity binders: a rigid cycle system reduces the conformational entropy penalty upon binding, stabilising the bound complex even when active-site mutations alter the pocket shape. However, the partial-correlation analysis shows this signal is confounded by molecular size; the relationship is therefore hypothesis-generating, with the size confound the primary alternative explanation.

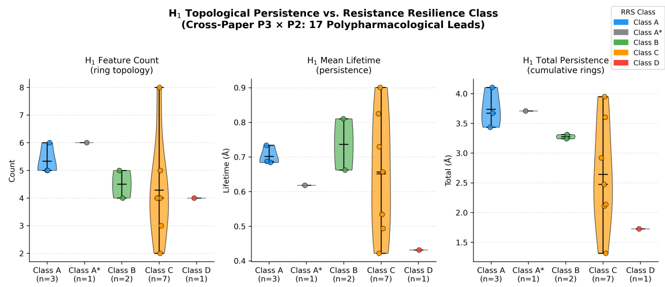

Figure 2: *Joint topological analysis linking TFP topological persistence features with molecular dynamics resistance classification (Temgoua et al., 2026b). Violin plots (pilot cohort, $n = 14$) show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Quantitative analysis of the expanded cohort ($n = 77$; Class A: $n = 46$, Class B: $n = 31$) demonstrates a nominally significant association between H_1 count and Resistance Resilience Score (RRS) across African-prevalent mutants (Spearman $\rho = 0.312$, $p = 0.0057$); the pilot $n = 14$ correlation ($\rho = 0.947$, $p < 0.0001$, on H_1 total persistence) is included for context but should be interpreted cautiously because the Class D subset contains only one compound. Caution is required when interpreting this association: after controlling for molecular weight the correlation collapses ($\rho_{partial} \approx 0$, $p > 0.7$), indicating a size-mediated effect rather than an independent topological signal.*

H_1 count could therefore serve as a computationally inexpensive, size-mediated surrogate for prioritising resistance-resilient candidates in future generative campaigns, pending larger size-controlled cohorts; dynamic persistent homology on full MD trajectories is planned for follow-up work.

4.8 Limitations

Several limitations constrain the interpretation of our results. First, TDA does not outperform ECFP4 overall in activity prediction (Table 1); TFP adds value primarily at the high-complexity tail of the distribution—molecules with ≥ 3 fused rings or ≥ 2 stereocentres. Second, activity labels are computational predictions from Ersilia ML models rather than experimental measurements, so all claims regarding "activity," "polypharmacology," and "resistance tolerance" should be understood as predictions within a computational framework, pending experimental validation. Third, the cross-paper H_1–RRS relationship ($\rho = 0.312$, $p = 0.0057$, $n = 77$; pilot $\rho = 0.947$, $n = 14$, on H_1 total persistence) is attenuated and, critically, does not survive control for molecular size (controlling for molecular weight collapses the correlation to $\rho_{\text{partial}} \approx 0$, $p = 0.85$): H_1 count should be interpreted as a proxy for molecular size rather than an independent topological predictor of clinical resilience. The quantum kernel circuit (canonical 6-qubit Phase-2 configuration) was evaluated on a classical simulator; we used the PennyLane `IQPEmbedding` template—a classically hard-to-simulate data encoding strategy—mitigating the limitations of simpler R_Y–R_Z alternating layers. The collapse of the quantum discriminator (AUC 0.425–0.511 versus 1.000) represents an inherent resolution bottleneck of the UMAP dimensionality reduction required for 8-qubit simulation: classical fingerprints excel at detecting local atom-level changes, while the quantum kernel remains suited for defining global applicability domains across heterogeneous spaces. The TFP reduces each persistence diagram to four summary statistics; the enriched TFP (32 features, adding persistence images and Betti curves) showed no improvement over the 12-feature TFP baseline (RF AUC 0.867 versus 0.867 at $n = 19\,849$; Table S12), consistent with the canonical full-library TFP AUC of 0.876. Finally, the library is biased toward the chemical space of its 396 African natural-product and 454 synthetic drug seeds; generalization to other chemical domains remains to be tested.

The QKS benchmark taught two methodological lessons. First, an apparent quantum advantage with the RBF gamma at its default value vanished once gamma was tuned by inner cross-validation (grid $\{0.5, 1.0, 2.0, 5.0\}$), and no deposited result file supports the earlier headline. Second, earlier 8-qubit runs reporting a quantum disadvantage ($n = 5\,000$: 0.752 versus 0.840, $p = 0.003$; $n = 19\,849$: 0.659 versus 0.825, $p = 0.0006$) were artefacts of the suboptimal circuit and a train/test scaling mismatch (C3); with the canonical 6-qubit circuit the quantum kernel shows no statistically significant difference from the gamma-tuned RBF kernel at any scale and is significantly better than linear at $n \geq 5\,000$ ($p \leq 0.0006$). These episodes underscore the importance of rigorous baseline tuning, circuit selection, and statistical power; QKS should therefore be positioned as a methodological, NISQ-era proof of concept rather than a predictive improvement.

Three contributions support the framework: (i) TFP resolves the scaffold paradox via H_1/H_0 decomposition, demonstrating that structural expansion preserves ring topology while varying substituent chemistry; (ii) TNE achieves 6.1× real-atom compression with competitive Tartarus regression ($R^2 = 0.473$ for PfDHFR); and (iii) H_1 count shows a nominally significant unadjusted association with resistance tolerance ($\rho = 0.312$, $p = 0.0057$, $n = 77$; pilot $\rho = 0.947$, $n = 14$, on H_1 total persistence; Figure 2) that the partial-correlation analysis reveals to be size-mediated—a cautionary example of size-confounding control. Classical ECFP4 (AUC 0.948) remains the recommended primary screening baseline; the canonical hybrid achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) but above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor (Table 1). We therefore position these quantum-inspired representations not as replacements for classical fingerprints but as complementary diagnostic tools that reveal topological dimensions of molecular structure.

5 Conclusion

We introduce three quantum-inspired molecular representations—the Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—and evaluate them on a 19 836-molecule canonical panel drawn from a 19 849-molecule library of antimalarial candidates derived from African natural-product chemical space. TFP, derived from Vietoris–Rips persistent homology, resolves the scaffold paradox (92.6 % Tanimoto distinction versus 69.3 % scaffold recovery) by demonstrating that H_1 ring topology is preserved (mean H_1 count 3.61) while H_0 connectivity diverges (mean H_0 count 38.04). TNE compresses the molecular tensor to a 192-element core descriptor (mean real ratio 6.1×; mean reconstruction error 0.113) while preserving activity-relevant information. The full computational protocol processed 19 849 molecules in 6.4 min (TDA) and 8.1 min (TNE, 488 s at 4 workers) on a standard workstation. The kernel benchmark with a properly tuned RBF baseline and the canonical 6-qubit circuit found no statistically significant difference between the quantum kernel and the gamma-tuned RBF kernel at any sample size ($n = 5\,000$: 0.820 versus 0.826, $p = 0.419$; $n = 19\,849$: 0.823 versus 0.829, $p = 0.060$, the latter a borderline trend toward lower QK performance), while the quantum kernel significantly outperformed the linear kernel at scale ($p \leq 0.000\,6$); earlier apparent quantum advantages and disadvantages were artefacts of untuned RBF hyperparameters and of the suboptimal 8-qubit circuit with a scaling mismatch. QKS is therefore best framed as a methodological framework contribution (a NISQ-era proof of concept) that shows no significant advantage over properly tuned classical kernels. The corrected classical-only benchmark confirms that classical ECFP4 (AUC 0.948) is the strongest single descriptor, while the canonical hybrid achieves AUC 0.888 ($p < 0.000\,1$ versus ECFP4), positioning quantum-inspired methods as effective complementary tools. All representations, benchmark results, and analysis scripts are deposited on Zenodo and GitHub under open-source licences.

Data Availability

All data supporting the conclusions of this study will be deposited on Zenodo (DOI: 10.5281/zenodo.19608875, reserved) and are mirrored in the project repository at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence; the deposit will be made available upon publication, and the repository is available to reviewers on request. The deposit includes: (1) TFP feature matrices for all 19 849 library molecules (H_0/H_1 persistence diagrams and summary statistics); (2) TNE embeddings (192-element Tucker core descriptors, bond dimension 8, reconstruction errors); (3) quantum kernel matrices (6-qubit IQPEmbedding, PennyLane 0.45.1); (4) all benchmark CSV files (5-fold CV splits, AUC scores per descriptor); (5) the cross-paper H_1/RRS analysis dataset; and (6) all analysis scripts.

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Authors' Contributions

Myke Vital Sao Temgoua: conceptualization, methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: conceptualization, methodology, software, supervision, writing – review & editing. Serge Guy Nana Engo: methodology, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. All authors read and approved the final manuscript.

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Ethics approval and consent to participate

Not applicable. This study is entirely computational and uses publicly available chemical databases and in-silico activity predictions; no human or animal subjects or patient data were involved.

Consent for publication

Not applicable. No individual person's data (including images, videos, or identifiable details) appear in this manuscript.

Competing Interests

The authors declare no competing interests.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants (Claude, Anthropic) to support code development, data analysis scripting, and manuscript drafting and editing. The authors reviewed and edited all AI-assisted output and take full responsibility for the content of this article. No generative AI tool is listed as an author. Computational results, statistical analyses, and all quantitative claims were generated, verified, and deposited by the authors.

References

D. Rogers and M. Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Maria Sorokina et al. COCONUT online: Collection of open natural products database. *Journal of Cheminformatics*, 13:2, 2021. doi: 10.1186/s13321-020-00478-9.

Jude Y. Betow, Ammar Qaseem, Boris D. Bekono, Yue Feng, Aurélien F. A. Moumbock, Conrad V. Simoben, Smith B. Babiaka, Solange A. Tanyi, Vanessa A. Shu, Ariane T. Ndi, Pascal Amoa Onguéné, Clovis S. Metuge, Simeon Akame, Donatus B. Eni, Cyril T. Namba-Nzanguim, Mathieu J. M. Tjegbe, Marianka N. L. Dikande, Patience T. Akeh, Sounders Balgah, Cyprian D. Meh, Akachukwu Ibezim, Idris F. Tabi, Yvette I. Malange, Bakoh Ndingkokhar, Leonel E. Njume, Kiran K. Telukunta, Said Amrani, Oyere T. Ebob, Wolfgang Sippl, Stefan Günther, and Fidele Ntie-Kang. African Natural Products Database (ANPDB): A resource for exploring the therapeutic potential of natural compounds from Africa. *Nucleic Acids Research*, 54(D1):D1336–D1344, January 2025. doi: 10.1093/nar/gkaf1186.

Godfrey Mayoka, Peter Mubanga Cheuka, Phanankosi Moyo, Godwin Akpeko Dziwornu, and Denzil Beukes. Value addition to African natural product-based drug discovery initiatives. *Journal of Natural Products*, 88(8):2018–2028, July 2025. doi: 10.1021/acs.jnatprod.5c00446.

JunJie Wee and Jian Jiang. A review of topological data analysis and topological deep learning in molecular sciences. *arXiv Preprint*, September 2025. URL https://arxiv.org/abs/2509.16877.

Yuxi (Jaden) Long and Bruce R. Donald. Interpretable binding affinity prediction with persistent homology. *PLOS Computational Biology*, 21(6):e1013216, 2025. doi: 10.1371/journal.pcbi.1013216.

Wenbo Zhang et al. Task-specific pre-training for molecular property prediction. *Briefings in Bioinformatics*, 27(1):bbag010, January 2026. doi: 10.1093/bib/bbag010.

Guillaume Simeon and Gianni de Fabritiis. TensorNet: Cartesian tensor representations for efficient learning of molecular potentials. In *Advances in Neural Information Processing Systems*, 2023. URL https://openreview.net/pdf?id=BEHlPdBZ2e.

Richard M. Milbradt, Shuo Sun, Christian B. Mendl, Johnnie Gray, and Garnet K.-L. Chan. Efficient application of tensor network operators to tensor network states. *arXiv Preprint*, January 2026. URL https://arxiv.org/abs/2601.19650.

Yousra Farhani. Quantum machine learning: Where are we and what is next? In *Proceedings of the 1st ACM SIGSPATIAL International Workshop on Quantum Spatial (Q-Spatial 2025)*, pages 15–18, 2025. doi: 10.1145/3764923.3777403.

Grier M. Jones, Viki Kumar Prasad, Ulrich Fekl, and Hans-Arno Jacobsen. Parametrized quantum circuit learning for quantum chemical applications. *Journal of Chemical Information and Modeling*, March 2026. doi: 10.1021/acs.jcim.6c00376.

Lohitaksh Badarala. Q-CaDD: accelerating in silico methodologies with quantum computation and machine learning for Epidermal growth factor receptor. *Scientific Reports*, 16:14436, March 2026. doi: 10.1038/s41598-026-44978-4.

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou, Wilfred Fon Mbacham, and Serge Guy Nana Engo. Computational discovery of antimalarial candidates from African natural product chemical space. *ChemRxiv*, 2026a. doi: 10.26434/chemrxiv.15006437/v1. Preprint, version v1 (posted July 2026).

John Preskill. Quantum computing in the nisq era and beyond. *Quantum*, 2:79, 2018. doi: 10.22331/q-2018-08-06-79.

K. Jacob, J. Clement, M. Arockiaraj, and A. Jindal. ChemGraphX: an open-source web tool for computing topological indices and entropy measures. *Journal of Cheminformatics*, 18:39, February 2026. doi: 10.1186/s13321-025-01141-x.

Stalin Arulsamy, Kiruthiga Natarajan, Mohankumar Ramar, and Parasuraman Pavadai. An interactive visualization platform for explainable cheminformatics: Universal Molecular Fingerprint Highlighter (UMFH). *In Silico Research in Biomedicine*, March 2026.

Andac Demir, Baris Coskunuzer, Ignacio Segovia-Dominguez, Yuzhou Chen, Yulia Gel, and Bulent Kiziltan. ToDD: Topological compound fingerprinting in computer-aided drug discovery. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 27978–27993, 2022. URL https://arxiv.org/abs/2211.03808.

Hui Cang and Guo-Wei Wei. Topologynet: Topology based deep convolutional and multi-task neural networks for biomolecular property predictions. *Journal of Chemical Information and Modeling*, 58(2):189–197, 2018. doi: 10.1021/acs.jcim.7b00661.

Soham Mukherjee, Shreyas N. Samaga, Cheng Xin, Steve Oudot, and Tamal K. Dey. D-gril: End-to-end topological learning with 2-parameter persistence. In *42nd International Symposium on Computational Geometry (SoCG)*, volume 367, pages 79:1–79:15, 2026. doi: 10.4230/LIPIcs.SoCG.2026.79.

Alejandro Giraldo, Daniel Ruiz, Mariano Caruso, Javier Mancilla, and Guido Bellomo. Q2sar: A quantum multiple kernel learning approach for drug discovery. *arXiv preprint*, 2025. URL https://arxiv.org/abs/2506.14920.

Rahul Khorana. Topological feature compression for molecular graph neural networks. *arXiv preprint*, 2025. URL https://arxiv.org/abs/2508.07807.

Zachary A. Rollins, Alan C. Cheng, and Essam Metwally. MolPROP: Molecular property prediction with multimodal language and graph fusion. *Journal of Cheminformatics*, 16(1):56, 2024. doi: 10.1186/s13321-024-00846-9.

Katarzyna Nałecz-Charkiewicz and Kamil Charkiewicz. Quantum computing in bioinformatics: A systematic review mapping. *Briefings in Bioinformatics*, 25(5):bbae391, 2024. doi: 10.1093/bib/bbae391.

Gautam Kumar, Sahil Yadav, Aniruddha Mukherjee, Vikas Hassija, and Mohsen Guizani. Recent advances in quantum computing for drug discovery and development. *IEEE Access*, 12: 64491–64509, 2024. doi: 10.1109/ACCESS.2024.3376408.

M. Lžičař and H. Gamouh. CHEESE: 3D shape and electrostatic virtual screening in a vector space. *ChemRxiv*, 2024. doi: 10.26434/chemrxiv-2024-cswth. URL https://doi.org/10.26434/chemrxiv-2024-cswth. Preprint.

David S. Wishart et al. DrugBank 5.0: a major update to the DrugBank database for 2018. *Nucleic Acids Research*, 46:D1074–D1082, 2018. doi: 10.1093/nar/gkx1037.

RDKit Contributors. RDKit: Open-source cheminformatics. https://www.rdkit.org/, 2024.

U. Bauer. Ripser: Efficient computation of vietoris–Rips persistence barcodes. *arXiv Preprint*, 2019. URL https://arxiv.org/abs/1908.02518.

J. Kossaifi, Y. Panagakis, A. Anandkumar, and M. Pantic. TensorLy: Tensor learning in Python. *Journal of Machine Learning Research*, 20(26):1–6, 2019.

Xanadu Quantum Technologies. PennyLane: Quantum machine learning library. https://pennylane.ai/, 2024.

Jan H Jensen. Augmenting genetic algorithms with deep neural networks for exploring the chemical space. *Chemical Science*, 10(12):3567–3572, 2019. doi: 10.1039/c8sc05372c.

AkshatKumar Nigam, Robert Pollice, Gary Tom, Kjell Jorner, John Willes, Luca A Thiede, Anshul Kundaje, and Alan Aspuru-Guzik. Tartarus: A benchmarking platform for realistic and practical inverse molecular design. In *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)*, 2023. URL https://arxiv.org/abs/2209.12487.

Asma Jamali, Tin Sum Cheng, and Rodrigo A. Vargas-Hernández. Spectral analysis of molecular features: When richer features do not guarantee better generalization. *arXiv preprint arXiv:2510.14217*, 2025. URL https://arxiv.org/abs/2510.14217. v2, June 2026.

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou, and Wilfred Fon Mbacham. Molecular dynamics validation of polypharmacological antimalarial leads targeting resistance mutations. *Journal of Chemical Information and Modeling*, 2026b. Companion manuscript (Paper 2), in preparation.